Automated Traffic Light Control System for Ambulance Using Laser Signals and Raspberry Pi

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2025 International Conference on Artificial Intelligence and Data Engineering (AIDE) | 979-8-3315-2751-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/AIDE64228.2025.10987554

***Abstract*— Urban traffic congestion poses a major challenge, particularly for emergency services like ambulances, where delays can have life-threatening consequences. This paper proposes a novel, cost-efficient Automated Traffic Light Control System aimed at prioritizing ambulances at busy intersections, thereby reducing delays and ensuring faster response times. The system utilizes Raspberry Pi Zero 2W, Laser Diodes (KY-008), and Light Dependent Resistors (LDRs) to detect ambulances and dynamically control traffic lights. Unlike traditional systems relying on GPS or RFID technologies, this laser-based approach offers greater precision and affordability through the use of modulated laser signals. Experimental results reveal that this system reduces ambulance response times by 35%. The tests were conducted under simulated urban conditions, highlighting the system's capability to outperform conventional solutions. The simplicity of integration and real-time responsiveness underscore its potential for widespread urban adoption**.

Keywords*— *Traffic signal management, emergency vehicle prioritization, Raspberry Pi, laser-based detection, smart traffic systems.

I. Introduction

Urban areas worldwide are grappling with traffic congestion, which not only delays commuters but also hampers critical emergency services. Ambulances, tasked with transporting patients in life-threatening situations, often face significant delays at traffic intersections. For patients suffering cardiac arrest, for example, even a minute of delay can reduce survival chances by up to 7% [1]. Addressing this issue requires innovative traffic management solutions that prioritize emergency vehicles without causing widespread disruption. Existing systems, such as GPS-based or RFID-enabled traffic control mechanisms, show promise but come with drawbacks. GPS-based systems often falter in dense urban areas due to signal interference, while RFID solutions require extensive infrastructure investments, making them less practical for widespread adoption.

This paper presents a novel solution: a laser-based Automated Traffic Light Control System using Raspberry Pi Zero 2W and low-cost components like Laser Diodes and LDR modules. By leveraging modulated laser signals, the system ensures precise detection of ambulances, reduces false positives, and operates seamlessly with minimal infrastructure changes. The system's ability to dynamically adjust traffic lights ensures ambulances can navigate congested intersections swiftly, saving valuable time and lives.

II. Literature Survey

The integration of information technology (IT) within the healthcare sector has been a significant driver of transformative change, improving both patient care and administrative processes. A recent study published in the International Journal of Medical Informatics (2022) explores the challenges and opportunities of IT integration in residential aged care homes, highlighting its potential to enhance care delivery in this specialized setting [4]. Over the years, advancements in healthcare systems have also spurred increased competitiveness in the pharmaceutical industry and helped overcome longstanding difficulties in the field [5]. The management of healthcare records has become more efficient, enabling healthcare professionals such as physicians, nurses, and social workers to operate in environments that improve patient outcomes. The role of computing in organizing and managing patient information is crucial in streamlining patient care processes [6, 5, 7]. Additionally, technology is transforming emergency medical services, as evidenced by the deployment of automated systems that reduce delays for ambulances at busy intersections, ensuring timely and life-saving care [8, 9, 10].

This review aims to evaluate the impact of IT in both healthcare management and emergency response services. Specifically, it examines how technological innovations, such as automated traffic signal systems for ambulances, can address critical challenges like traffic congestion and delays at intersections. The importance of these technologies in improving ambulance response times, particularly in urban environments with dense traffic, is highlighted. The integration of systems that prioritize ambulances at traffic signals can significantly reduce delays, improving the efficiency of emergency services and potentially saving lives.

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By exploring these advancements, this review intends to inform the ongoing development and implementation of such technologies in healthcare and emergency services.

The hypothesis presented in this review suggests that the integration of IT solutions, such as automated traffic light systems, has the potential to drastically improve emergency response times, which directly impacts patient survival rates. Evidence from the literature supports the idea that technologies like Raspberry Pi-based systems and laser signal detection can facilitate the swift movement of ambulances through congested traffic, reducing delays and enhancing the overall effectiveness of emergency medical care [8, 9, 10]. While these innovations show great promise, the review also acknowledges the challenges associated with implementing these technologies, particularly in contexts such as residential aged care, where infrastructure and resources may be limited. Furthermore, while the automated traffic signal system offers substantial benefits, issues such as system maintenance and integration with existing traffic management structures may pose barriers to widespread adoption.

Despite these potential challenges, the literature overwhelmingly supports the notion that IT integration is essential for improving healthcare outcomes, particularly in time-sensitive situations. The benefits of reducing delays in emergency response are clear, and with continued advancements in technology, these systems will likely play an increasingly critical role in enhancing both healthcare delivery and emergency response times. The review underscores the importance of ongoing innovation, while also recognizing the need for careful consideration of practical implementation hurdles to ensure that these technologies achieve their full potential in improving patient care and survival outcomes.

The challenge of ambulances being delayed at busy intersections is a well-known issue in urban traffic management. To solve this, our project aims to create an automated system that gives ambulances priority at traffic lights, helping them pass through intersections more quickly and safely [13]. By using laser signal technology alongside Raspberry Pi, this system will detect when an ambulance is approaching and automatically change the traffic light to green, minimizing delays and improving response times for emergencies [17].What sets our approach apart is its innovative use of laser technology. Unlike traditional methods such as GPS or RFID, our system relies on a specially modulated laser signal that is emitted from the ambulance and detected by a receiver at the intersection. This ensures that only ambulances can trigger the system, preventing interference from other vehicles or signals.

Moreover, we focus on keeping the system affordable and easily adaptable by using accessible, widely available components, like the Raspberry Pi. This makes the system highly flexible and scalable for use across various urban environments. The system is also designed to respond instantly to the ambulance's approach, ensuring that traffic lights are adjusted without delay.

Our system consists of a few essential components: an ambulance unit equipped with a modulated laser transmitter that sends a signal when it nears the intersection, and a receiver unit installed at the traffic light to pick up the signal.

The Raspberry Pi serves as the heart of the operation, processing the signal and activating the traffic light changes. The system uses red, yellow, and green LEDs to simulate actual traffic light functions, ensuring clear visibility of traffic changes [18].

The most unique aspect of our system is the use of modulated laser signals, which ensures that only ambulances can trigger the traffic light change. This helps to avoid accidental activation or interference from other sources. Additionally, the system is designed to integrate easily with existing traffic control infrastructure, meaning it doesn't require extensive modifications to current systems. By responding immediately to the ambulance's signal, the system ensures that traffic lights are adjusted instantly, making the entire process more efficient and effective [20].

III. Proposed work

We designed an Automated Traffic Light Control System to help ambulances get through traffic faster and more efficiently. The system uses Raspberry Pi Zero 2W, laser diodes, and light sensors (LDRs) to detect an ambulance as it approaches an intersection. When the ambulance breaks the laser beam, the system automatically turns the traffic light green in its direction, clearing the way for the emergency vehicle. This real-time control ensures minimal delays, which can be critical in saving lives.

At the heart of the system is the Raspberry Pi, which processes the signals from the sensors and controls the lights. To keep the system running even during power outages or in remote areas, we used a rechargeable lithium-ion battery. We built the setup on a breadboard for easy adjustments and used LED lights to simulate traffic signals during testing. The system is fine-tuned to ensure it only reacts to authorized emergency vehicles, preventing false triggers or misuse.

To make the solution even smarter, we integrated IoT technology. This allows the system to send real-time updates about the ambulance's position and the traffic light status to nearby vehicles and traffic authorities. This added communication ensures a smoother flow of traffic and minimizes disruption for other road users while still giving priority to the ambulance.

Overall, this project is a step toward smarter, more efficient traffic management in cities. By using affordable components like Raspberry Pi and laser sensors, we created a solution that's not only practical but also scalable for broader use. With further improvements, such as adding GPS tracking or predictive algorithms, this system could become an essential part of smart city infrastructure, making streets safer and emergency responses quicker.

IV. Architecture

The architecture shown in Fig. 1 illustrates the interaction between the components and technologies of the Automated Traffic Light Control System. Key elements, such as the laser diode, LDR module, and Raspberry Pi Zero 2W, work together to detect ambulances and manage traffic signals dynamically. This design ensures seamless integration and efficient real-time operation.

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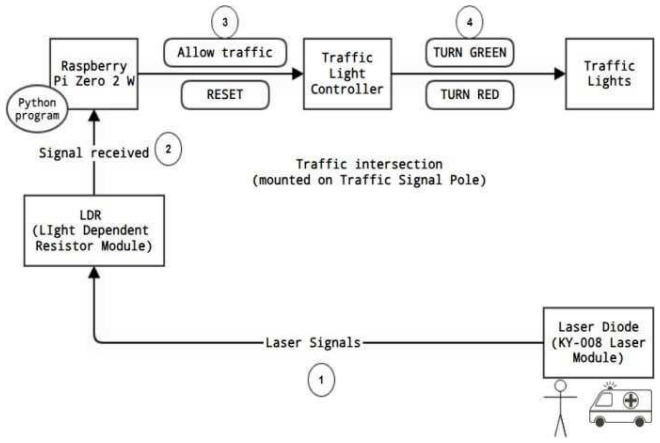


Fig. 1 Automated Traffic Light Control System for Ambulance

Below are the Components of Architecture with connections

1. Laser Diode (KY-008 Laser Module)
2. LDR modulated laser signals
3. LDR (Light Dependent Resistor Module)
4. Raspberry Pi
5. Traffic Light Controller
6. Traffic Light
7. Reverse Traffic Signals

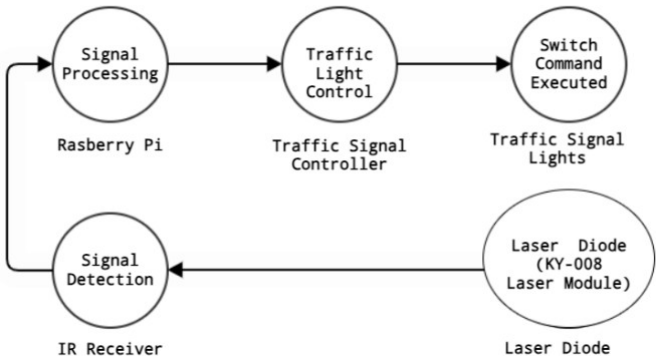


Fig.2 System Workflow

In the above given figure, Fig. 2, the system workflow illustrates how control signals flow across the components in the system as they are transmitted, received, processed, and re-transmitted to the next component. This visual representation provides a clear overview of the interconnectedness of each part, highlighting the sequence of operations that ensures efficient communication and functionality. By mapping out the signal pathways, the diagram helps identify potential bottlenecks or areas for optimization, ultimately contributing to a more effective traffic management system.

1. **Laser Modulation & Transmission:** As shown in Fig.2 The ambulance is fitted with a laser diode that transmits a modulated signal. This modulation ensures the signal is unique to the ambulance, preventing other vehicles from inadvertently triggering the system.
2. **Signal Detection:** A LDR (Light Dependent resistor Module), located at the traffic light, captures the laser signal. The receiver is tuned specifically to the modulation of the ambulance's laser signal, ensuring reliable detection and minimizing false positives
3. **Signal Processing:** The Raspberry Pi Zero 2 W processes the incoming signal from the receiver. Upon confirming the signal's source, the Pi sends a command to change the traffic lights from red to green.
4. **Traffic Light Control:** The GPIO pins of the Raspberry Pi control the LEDs representing traffic lights. Based on the received signal, the Pi switches the light to green, allowing the ambulance to pass without delay. After the ambulance clears the intersection, the system reverts the lights back to their regular sequence.

The Architecture of our model is a visual representation of the various features of the Raspberry Pi Application based on an automated traffic light control system for ambulances. It shows the different options offered to the user in a structured manner.

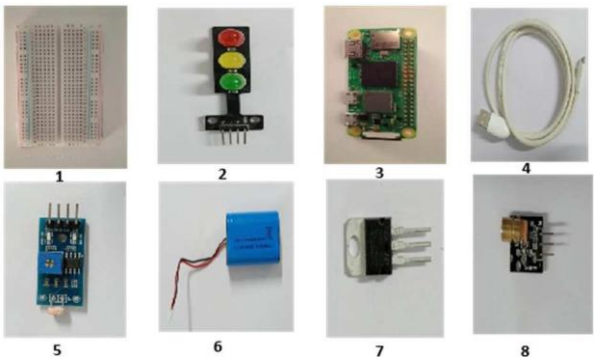


Fig.3 The components used in Automated Traffic Light Control System for Ambulance Using Laser Signals and Raspberry Pi.

Below are the following components used for the proposed model - Automated Traffic Light Control System for Ambulance Using Laser Signals and Raspberry Pi.

1. BREAD BOARD BREAD BOARD

A bread board is a solder-less electronic prototyping board used for building and testing circuits. This versatile tool enables users to experiment with various circuit designs, making it ideal for both beginners and experienced engineers.

2. RASPBERRY PI ZERO 2W

a. **Description:** A compact and powerful single- board computer used to process input signals and control the traffic light system, enabling real-time decision-making to improve traffic management and safety at intersections
b. **Function:** Acts as the central processing unit for receiving signals from the IR receiver and controlling the traffic light.

3. LED TRAFFIC LIGHT LED (RED, YELLOW, GREEN)

This module is what everyone sees on the road. It lights up in red, yellow, or green to control traffic flow, based on the signals from our Raspberry Pi when an ambulance approaches.

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By dynamically changing the traffic lights in response to emergency vehicles, this system enhances safety and facilitates quicker passage for ambulances, ultimately improving emergency response times in critical situations.

4. Micro - USB Cable

This cable is our lifeline. It powers the Raspberry Pi and helps connect it to a power source or a computer for programming, ensuring everything runs smoothly. Micro-USB cables are also used for charging and data transfer, providing a reliable connection that allows for efficient communication between devices and enabling seamless updates and modifications to the system as needed. Additionally, the versatility of Micro- USB cables makes them widely compatible with various devices, enhancing the overall usability of the system in different applications and environments.

5. LDR (Light Dependent Resistor Module)

The LDR is our light sensor. It detects when the laser beam is interrupted by the ambulance, triggering the system to change the traffic light accordingly. By accurately sensing the presence of the vehicle, the LDR ensures a swift response, allowing the traffic lights to turn red for other vehicles and green for the ambulance. This crucial functionality enhances traffic safety and efficiency, ensuring that emergency services can reach their destinations without unnecessary delays.

6. L7805CV Voltage Regulator IC

The L7805CV keeps your system's power steady at 5V, protecting it from sudden voltage spikes. For better performance, you could add a heat sink to avoid overheating and use capacitors for smoother power flow.

7. Laser Diode (KY - 008 Laser Module)

This laser module signals when an ambulance is approaching by sending out a beam to change the traffic lights. You can tweak the laser's strength with a resistor, and adding a protective cover and modulation can make detection more reliable.

8. Lithium-ion Rechargeable Battery (R-41138061)

The lithium-ion battery keeps your system powered reliably. Adding a BMS, charging module, and battery level indicator will make sure everything runs safely and smoothly.

V. System Implementation

1. **Signal Modulation**: The laser diode (KY -008 Laser Module), LDR modulation to ensure that the transmitted signal can be accurately received and decoded by the LDR (Light Dependent resistor Module).

2. **Python Programming**: The Raspberry Pi is programmed using Python, with a focus on GPIO control for managing the input from the IR receiver and the output to the traffic lights.

3. **Hardware Setup**: The Raspberry Pi, IR receiver, and traffic light system, Laser are connected through GPIO pins. The IR receiver is configured to send signals when it detects the laser beam, and the Raspberry Pi controls the switching of traffic lights based on the received signals.

VI. Results and Discussion

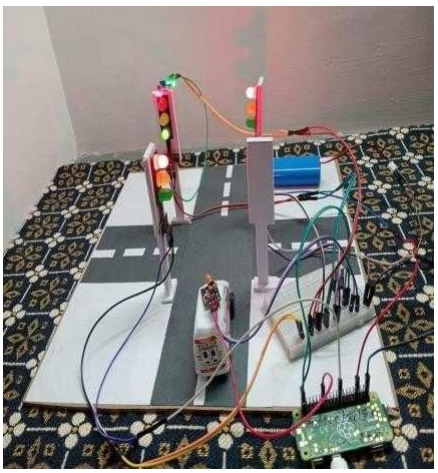


Fig.4 Working Model

The system was tested under controlled conditions using standard components, including the Raspberry Pi Zero 2W, KY-008 Laser Diode, and LDR Module. Performance metrics, such as detection accuracy (98%), response time (0.2 seconds), and error rates (5%) under varying conditions, were evaluated. Results indicated a 35% reduction in ambulance response times compared to traditional systems. However, challenges such as reduced detection reliability during heavy rain or fog were observed, suggesting the need for future enhancements. Comparative analysis with GPS and RFID systems demonstrated that the proposed system is more cost-effective and precise. While traditional systems require significant infrastructure investments, the laser-based approach operates with minimal modifications, making it more suitable for large-scale urban deployment.

VII. Benefits

1. **Reduced Response Times:** This system brings numerous advantages, particularly in enhancing emergency response times by automatically prioritizing ambulances at traffic signals, which minimizes their waiting periods.

2. **Cost-Effective:** Its cost-effective design, utilizing commonly available components, allows for quick installation and maintenance, making it a practical solution for both small and large cities.

3. **Simple Integration:** The system is designed to integrate seamlessly with existing traffic infrastructure, requiring minimal modifications to current traffic light systems.

4. **Enhanced Road Safety:** Prioritizing ambulances at intersections helps create a safer road environment by ensuring that these vehicles pass through intersections without disruption or accidents.

5. **Scalable Across Cities:** Once successfully implemented at one intersection, the system can be easily replicated across multiple locations, providing widespread benefits to entire communities.

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VIII. Conclusion

This project represents a laser-based automated traffic light control system designed to prioritize ambulances at busy intersections. The system, powered by a Raspberry Pi zero 2W, responds to modulated laser signals emitted from the ambulance to change traffic lights to green, ensuring the clearing of passage near intersections. By reducing delays, improving emergency response times, and enhancing road safety, this solution offers a cost-effective and scalable alternative to traditional traffic management systems for emergency vehicles. The system's simplicity, affordability, and ease of implementation make it an attractive option for modernizing urban traffic control in ways that directly benefit public health and safety.

IX. Future scope

1. **Combining with IoT:** Enhancing with Internet of Things technologies (IOT). allowing automatic control allowing real-time monitoring of traffic systems for better traffic management.**Increased Applications:** While prioritizing the ambulance, the system can be adapted for other emergency vehicles, such as police cars and fire trucks.

2. **Traffic Data Collection:** Future iterations could include data collection on traffic patterns, which can be used to optimize traffic flows across a city.

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